Weibull–Hyperbola Model

User Manual

Version 8

Interactive Web Application for Fatigue Data Analysis

Application type:	Single-page web application
Models:	M0–M7 (8 nested models)
Estimation:	Maximum likelihood + grid search
Simulation:	Physical-branch guaranteed
Exports:	CSV, LaTeX, PDF, PNG

This document describes how to use the Weibull–Hyperbola Model web application for fitting, simulating, and analysing fatigue life data.

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1 Introduction

The **Weibull–Hyperbola Model** is a browser-based tool for analysing fatigue data. It combines a hyperbolic stress–life relationship with a three-parameter WEIBULL distribution to model the scatter in fatigue life.

The application allows you to:

- Simulate synthetic fatigue data on a guaranteed physical branch.
- Fit up to eight nested models (M0–M7) to experimental data.
- Estimate reference parameters σ_{M0} and N_0 automatically.
- Compare models using AIC and BIC.
- Visualise percentile curves in both transformed and original coordinates.
- Export results as CSV, LaTeX tables, PDF and PNG figures.

No installation required

The application is a single HTML file. It runs entirely in your browser — no server, no installation, no data leaves your computer. All computation is performed locally in JavaScript.

2 The Mathematical Model

2.1 Transformed variables

The model uses three dimensionless variables:

$$\pi_1 = \frac{\sigma_m}{\sigma_M} \qquad \text{(stress ratio)} \tag{1}$$

$$\pi_2 = \frac{\sigma_M}{\sigma_{M0}} - 1 \qquad \text{(normalised amplitude stress)} \tag{2}$$

$$\pi_3 = \log_{10}\left(\frac{N}{N_0}\right) \qquad \text{(normalised log-life)} \tag{3}$$

where σ_m is the mean stress, σ_M is the stress amplitude, N is the number of cycles to failure, and σ_{M0}, N_0 are reference values.

2.2 The hyperbola

The core relationship is:

$$V = w_1\pi_1 + w_{12}\pi_1\pi_2 + w_{13}\pi_1\pi_3 + \pi_2\pi_3 + w_{123}\pi_1\pi_2\pi_3 \tag{4}$$

Solving for π_2 as a function of π_3 at fixed π_1 gives a hyperbola:

$$\pi_2 = \frac{V - w_1\pi_1 - w_{13}\pi_1\pi_3}{\pi_3 + w_{12}\pi_1 + w_{123}\pi_1\pi_3} \tag{5}$$

The vertical asymptote is at $\pi_3^{\text{asym}} = -w_{12}\pi_1/(1+w_{123}\pi_1)$ and the horizontal asymptote at $\pi_2^{\text{asym}} = -w_{13}\pi_1/(1+w_{123}\pi_1)$.

2.3 The Weibull distribution

The quantity V follows a three-parameter WEIBULL distribution:

$$f(v) = \frac{\beta}{\delta}\left(\frac{v-\lambda}{\delta}\right)^{\beta-1}\exp\left[-\left(\frac{v-\lambda}{\delta}\right)^{\beta}\right], \quad v > \lambda \tag{6}$$

with location λ, scale δ, and shape β.

2.4 The eight nested models

Model	Free parameters
M0	$w_1, w_{12}, w_{13}, w_{123}$
M1	w_1, w_{12}, w_{13}
M2	w_1, w_{13}, w_{123}
M3	w_1, w_{12}, w_{123}
M4	w_1, w_{123}
M5	w_1, w_{13}
M6	w_1, w_{12}
M7	w_1

Parameters not listed are fixed to zero. All models are nested within M0.

3 Getting Started

3.1 Opening the application

Open the HTML file in any modern browser (Chrome, Firefox, Edge, Safari). No installation or internet connection is required after the first load, except for the Chart.js and jsPDF libraries which are loaded from local files.

3.2 Interface overview

The interface is organised into collapsible sections:

Section	Purpose
Model & Parameters	Choose the model, edit w parameters, Weibull parameters, and references.
Simulation Settings	Define the simulation ranges and run it.
Data Extrema	Automatic read-only display of data ranges.
Fitting & Estimation	Load CSV, estimate references, fit models.
Results	Tables, charts, and export options.
Console Log	Messages, warnings, and progress information.

Click any section header to expand or collapse it. The top-right button **Expand all / Collapse all** toggles all sections at once.

3.3 The status indicator

A coloured dot in the top-right corner of the header shows the application state:

- **Green** — ready / idle.
- **Red** — computing. Do not close the tab.

3.4 The Case / Title field

The text field in the header is used as the **filename prefix** for all exports. If you load a CSV file, it is set automatically from the filename.

4 Workflow 1: Simulating Data

This workflow generates synthetic data from known parameters.

4.1 Step 1 — Set the model and parameters

1. Expand **Model & Parameters**.
2. Choose a model from the *Model* selector (e.g., M0). The free w fields update automatically.
3. Enter the w values. A good starting set is $w_1 = 0.5$, $w_{12} = -0.2$, $w_{13} = 0.3$, $w_{123} = -0.1$.
4. Enter the WEIBULL parameters: $\lambda = 0.5$, $\delta = 0.3$, $\beta = 2.5$ (defaults).
5. Enter the references: $\sigma_{M0} = 40$, $N_0 = 1000$ (defaults).

Reference constraints

σ_{M0} must be strictly less than $\sigma_M^{\min}$ in your simulation range. N_0 must be less than the runout threshold. The application auto-corrects violations and reports them in the console.

4.2 Step 2 — Set the simulation ranges

Expand **Simulation Settings** and set:

- $\sigma_M^{\min}$, $\sigma_M^{\max}$ — amplitude stress range (e.g., 50–200 MPa).
- $\sigma_m^{\min}$, $\sigma_m^{\max}$ — mean stress range (e.g., 10–150 MPa).
- **N points** — number of points to generate (default 1000).
- **Runout threshold** — cycles above which a point is censored (default 10^7).
- **Apply runout censoring** — if checked, points above the threshold are marked as runouts.

4.3 Step 3 — Simulate

Click **>Simulate Data**. The console reports:

- Number of valid points generated.
- Rejection rate (percentage of attempts rejected by physical constraints).

Reading the rejection rate

A rejection rate below 80% is normal. Above 80% suggests your w parameters, ranges, or references are poorly matched. The console will warn you.

4.4 Step 4 — Plot the curves

Click **[Plot] Plot Curves (sim.)** to display the percentile curves. This uses the parameters currently in the **Model & Parameters** section (it does *not* fit anything).

5 Workflow 2: Fitting Experimental Data

This workflow estimates parameters from measured fatigue data.

5.1 Step 1 — Prepare your CSV file

Your file must be comma-separated with the following required columns:

Column	Description	Units
`sigmam`	Mean stress σ_m	MPa
`sigmaM`	Stress amplitude σ_M	MPa
`N`	Cycles to failure	cycles

An optional `runout` column (0/1) can be included. If absent, it is inferred from the runout threshold.

Example:

```
sigmam,sigmaM,N
20,180,15000
30,160,42000
40,140,98000
...
```

5.2 Step 2 — Load the file

Either:

- Drag the file onto the drop zone, or
- Click the drop zone to browse for the file.

Upon loading, the application:

1. Clears all previous results.
2. Sets the *Model* selector to **All** and checks all model checkboxes.
3. Updates the **Data Extrema** section with the min/max of each variable.
4. Sets the **Case / Title** from the filename.
5. Displays the data table in the **Results** tab.

5.3 Step 3 — Configure the fitting

Expand **Fitting & Estimation**:

1. **Models to fit** — tick the models to include (all ticked by default).
2. **Estimation method**:
   - **Option 1 (recommended)** — each model independently estimates σ_{M0} and N_0.
   - **Option 2** — joint estimation: one σ_{M0}/N_0 pair for all models, then re-fit.
3. **Grid search configuration**:
   - **Grid points** (3–20) — resolution of the initial coarse grid.
   - **Grid restarts** (1–15) — L-BFGS restarts during the grid phase.
   - **Refinement restarts** (3–30) — L-BFGS restarts during the final refinement.

Performance tip

Default values (3, 3, 5) are fast. For publication-quality fits, use 5–7 grid points and 5–10 restarts. Larger values take proportionally longer.

5.4 Step 4 — Estimate or fit

Two buttons are available:

- **[Plot] Estimate σ_{M0} & N_0** — runs the full grid search + refinement.
- **[Fit] Fit (manual σ_{M0}, N_0)** — fits models at the reference values currently entered in **Model & Parameters**, skipping the search.

During execution, the status dot turns red. Progress is shown in the progress bar and the console.

5.5 Step 5 — Review results

Once fitting completes, the **Results** section becomes visible with five tabs:

Tab	Content
Data	Loaded experimental data (paginated).
Model Comparison	Parameter estimates, AIC, BIC for all fitted models.
Percentile Curves	Interactive charts (see Section 6.3).
Simulated Data	Table of simulated data (if any).
Export	CSV, LaTeX, PDF and PNG download buttons.

The best model (lowest AIC) is highlighted with a green background and a ***** in the model comparison table. Its equation is shown above the tabs.

6 The Results Tabs in Detail

6.1 Data tab

Shows the loaded experimental data with an additional column $R = \sigma_m/\sigma_M$ (stress ratio). Tables are paginated at 10 rows per page.

6.2 Model Comparison tab

Displays one row per fitted model with columns:

Column	Meaning
*	Marks the best model (lowest AIC)
Model	Model identifier M0–M7
k	Number of free parameters
σ_{M0}, N_0	Estimated references
$w_1, w_{12}, w_{13}, w_{123}$	Estimated w parameters
λ, δ, β	Weibull parameters
ℓ	Log-likelihood
AIC, BIC	Information criteria

6.3 Percentile Curves tab

Shows the fitted model as a family of percentile curves ($p = 0.01, 0.05, 0.25, 0.50, 0.75, 0.95, 0.99$) at six values of π_1.

Coordinate systems

Use the toggle in **Model & Parameters** to switch between:

- π_2 **vs** π_3 — transformed coordinates.
- σ_M **vs** N — original physical coordinates (default).

Opacity slider

The slider above the charts controls how the scatter points fade with distance from the slice π_1 value:

- **0** — all points equally opaque.
- **4.6** — points at the median distance are nearly invisible; points near the slice stay bright.

This helps visualise which data points belong to each π_1 slice.

Scrolling

The charts are laid out horizontally. Scroll sideways (or swipe on touch devices) to see all slices.

Overlaid data

When **Include simulated data** is active (see Section 8), the simulated points are shown in blue and the real experimental data in red.

6.4 Simulated Data tab

Shows the generated synthetic data (when simulation was run). Paginated like the other tables.

6.5 Export tab

Offers:

- **[Download] Data CSV** — experimental data as CSV.
- **[Download] Models CSV** — parameter estimates as CSV.
- **[Download] Data LaTeX** — experimental data as a booktabs table.
- **[Download] Models LaTeX** — parameter table as booktabs.
- **[Download] Simulated CSV** — simulated data as CSV.
- **[Plot] Export Figures (PDF)** — one PDF per chart, A4 landscape.
- **[Plot] Export Figures (PNG)** — one PNG per chart, 3× resolution.

All filenames use the **Case / Title** as prefix. Below the buttons, thumbnails of each figure appear; clicking a thumbnail downloads that figure as PNG.

7 Reference Parameters and Auto-Correction

7.1 Constraints

- σ_{M0} must be **strictly less** than the minimum σ_M in the data (otherwise $\pi_2 \leq 0$).
- N_0 must be **strictly less** than the minimum N in the data (otherwise $\pi_3 \leq 0$).

7.2 Auto-correction

When you click **Simulate Data** or **Fit (manual)**, if either reference violates its constraint, the application automatically corrects it to 0.9× the corresponding minimum and writes a warning to the console.

When you click **Estimate σ_{M0} & N_0**, the search ranges are automatically chosen inside the feasible region.

Why does this matter?

The hyperbola has two branches. Only one branch gives physically valid fatigue data (positive σ_M, positive N). The application enforces a constraint $\pi_3 + w_{12}\pi_1 + w_{123}\pi_1\pi_3 > 0$ for all data points, which guarantees the correct branch.

8 Including Simulated Data in Curves

The toggle **Include simulated data** (in Simulation Settings) has three effects:

1. **After a fit** — a new simulation is generated using the fitted parameters. The curves are drawn from the simulated points, and the real data is overlaid in red.
2. **After Plot Curves (sim.)** — the curves are drawn from the last simulation, and (if a real dataset is loaded) the real data is overlaid in red.
3. **When off** — the curves use only the simulated or fitted data, without overlay.

Use case

This is useful when your experimental dataset is small: the percentile curves are better defined by a dense simulation matched to the fitted parameters.

9 Resetting the Application

The button **[Warning] Reset and load file** (in **Fitting & Estimation**) performs a full reset:

1. Shows a confirmation dialog.
2. Clears all data, results, charts, and settings.
3. Reloads the page.
4. Opens the file dialog for a fresh CSV.

Irreversible

This action cannot be undone. All unsaved work (including figures you have not exported) is lost.

10 Reading the Console

The console (bottom of the page) uses colour coding:

Colour	Meaning
Green **[OK]**	Success, completion.
Orange **[Warning]**	Warning — non-fatal issue.
Red **[Error]**	Error — action failed.
Blue	Informational message.
Gold bold	Section header / start of operation.

Important messages to watch for:

- *"Reference auto-corrected to ..."* — your reference was infeasible; the application fixed it.
- *"Rejection rate: XX%"* — percentage of simulation attempts rejected by physical constraints.
- *"Only XX/YYY valid points"* — simulation could not reach the requested count.
- *"Best model: Mn"* — best fit by AIC.

11 Troubleshooting

"No valid points were generated"
The simulation constraints are too tight. Reduce $\sigma_M^{\min}$ closer to σ_{M0}, widen the ranges, or check that $\sigma_{M0} < \sigma_M^{\min}$.

"CSV parse error"
Verify the header contains `sigmam`, `sigmaM`, `N`. Check that no cells are empty or contain non-numeric characters.

"Fit did not converge"
Try more restarts (e.g., 10 grid, 10 refine), or manually enter plausible σ_{M0} / N_0 and use **Fit (manual)**.

Curves are missing or truncated
The plotted range may be too narrow. Increase the simulation range or extend the data.

"Only 1 valid π_1 slice" in console
The data does not admit multiple slices that satisfy all percentile curves in the positive quadrant. Widen the w parameters or review the fit.

PDF export does nothing
Check that the jsPDF library is available (network access to the CDN or local file present). The console will show an error if not.

12 Quick Reference

Button	Action
>Simulate Data	Generate synthetic points from the current parameters.
[**Plot**] Plot Curves (sim.)	Draw percentile curves from the simulation parameters.
[**Plot**] Estimate σ_{M0} & N_0	Grid search + refinement across all selected models.
[**Fit**] Fit (manual)	Fit at the manually entered references.
[**Warning**] Reset and load file	Full reset and load a fresh CSV.
[**Download**] CSV / LaTeX	Download tables.
[**Plot**] PDF / [**Plot**] PNG	Download figures.

Recommended workflow

1. Load your CSV.
2. Estimate σ_{M0} & N_0 (Option 1, defaults).
3. Inspect the Model Comparison tab; note the best model.
4. Review the Percentile Curves; adjust the opacity slider if needed.
5. Tick *Include simulated data* and re-plot for a denser curve family.
6. Export figures and tables for reporting.

Version History

v8 — Physical-branch guarantee in simulation; rejection-rate reporting; per-model independent estimation (Option 1); per-figure PDF export; export previews; strict π_1-slice validation for percentile curves; opacity slider; model equation display.